Fundamental Trigonometric Identities

Using the Pythagorean identities

1 Given that $\sin\theta = \frac{3}{5}$ and θ lies in Quadrant II, find:

a $\cos\theta$ **b** $\tan\theta$ **c** $\sec\theta$ **d** $\csc\theta$

2 Given that $\tan\theta = 2$ and θ lies in Quadrant III, find $\sin\theta$ and $\cos\theta$ in exact simplified form.

Simplifying and verifying

3 Simplify each expression to a single trigonometric function:

a $\frac{1-\cos^2 x}{\sin x}$

b $\sec x \cot x$

c $\tan x \csc x$

4 Verify the identity $(1+\tan^2 x)\cos^2 x = 1$.

5 Verify the identity $\frac{\sin x}{1+\cos x} = \frac{1-\cos x}{\sin x}$.

Quotient and reciprocal identities

6 Rewrite each expression using only $\sin x$ and $\cos x$, then simplify:

a $\cot x \sin x$

b $\frac{\sec x}{\csc x}$

c $1-(\sin^2 x \csc^2 x)\sin^2 x$

7 Simplify $\frac{\cos x}{1-\sin x} - \tan x$ to a single term.

More identity verifications

8 Verify the identity $\sec^2 x - \tan^2 x = 1$ using the Pythagorean identity for tangent and secant.

9 Verify the identity $\frac{1}{1-\cos x} + \frac{1}{1+\cos x} = 2\csc^2 x$.

10 Show that $(\sin x + \cos x)^2 = 1 + 2\sin x \cos x$.

11 Given that $\cos\theta = -\frac{8}{17}$ and θ lies in Quadrant III, find $\sin\theta$, $\tan\theta$, and $\cot\theta$ in exact simplified form.

Identities with sums and products

12 Verify the identity $\sin x \cos x(\tan x + \cot x) = 1$.

13 Simplify $(\sec x - \tan x)(\sec x + \tan x)$ to a constant.